

GravField: Entangling Digital Objects with Bodies

Botao Amber Hu ^{id *}
Reality Design Lab
New York City, USA

Yuemin Huang ^{id †}
East China Normal University
Shanghai, China

Mingze Chai ^{id ‡}
Independent
Shanghai, China

Xiaobo Aaron Hu ^{id §}
Independent
Shanghai, China

Yilan Elan Tao ^{id ¶}
Reality Design Lab
New York City, USA

Rem Rungu Lin ^{id ||}
The Hong Kong University of
Science and Technology (Guangzhou)
Guangzhou, China



ABSTRACT

GravField ("Gravitational Field") is a live-coding and intercorporeal improvisation performance in a collocated Mixed Reality (MR) environment. The project investigates how digital objects mediate intercorporeal movement and how participants internalize and adapt to a new "digital physics" to form collective behaviors. Through MR environments where digital elements shape both cognitive and physical experiences, we combine physical performance with MR to examine the dynamic relationships between bodies, digital objects, and live musical feedback.

Index Terms: Augmented Reality, Collocated Mixed Reality, Live Coding, Dance, Affordance, Social Play, Intercorporeality, Contact

*e-mail: botao@reality.design

†e-mail: uminh1024@gmail.com

‡e-mail: chaimignze@gmail.com

§e-mail: agalloch21@gmail.com

¶e-mail: elan@reality.design

||e-mail: rlin408@connect.hkust-gz.edu.cn

Improvisation, Movement-Sound Interaction

1 PROJECT URL

<https://reality.design/project/gravfield>

2 DESCRIPTION OF ARTWORK

2.1 Thematic Statement

In a future dominated by mixed reality where technology fades into the background of our lives, it becomes increasingly challenging to distinguish the physicality of digital objects. These digital objects wield the power to reshape our collective social reality. Operating under 'digital physics', these objects influence how our bodies respond, move, and interact. In this artwork, we want to explore how digital objects mediate our intercorporeal movement. How will people gradually internalize this new physics and adjust to form collective behavior?

2.2 Artwork

GravField ("Gravitational Field") is a live-coding intercorporeal improvisation performance set in a collocated Mixed Reality (MR) environment. Drawing inspiration from contact improvisation techniques, this performance explores the dynamic interplay between

bodies, mediated through live-codable MR Audio-Visual Virtual Mediators (AVVMs). These AVVMs employ physical metaphors such as "Springs," "Ropes," and "Magnetic Fields" to facilitate relationships among participants wearing MR head-mounted displays (HMDs) and a live musician. The landscape of these mediators, which we term "digital physics," is dynamically adjusted by live coders, actively influencing participants' collective movements, which in turn shape the live music generation. Our performance aims to reveal how AVVMs blend with participants' intercorporeal movements, fostering emergent coordination and interdependent behaviors. Rooted in postphenomenology, GravField illuminates the potential for future more-than-human entanglement assemblages and interplay systems in holistic interactive MR environments.

2.3 The Performance

Immersant's Experience

When the performance begins, dancers are invited to wear the MR HMD HoloKit, transforming into "immersants" in the process.

Through the stereoscopic MR display, all immersants perceive themselves as connected by AVVMs. These AVVMs assume a crucial role in our audio-visual system, providing diverse possibilities through their various forms. As the immersants move, the live-coder dynamically generates the AVVM shapes and corresponding sound effects in response to their movements. These audio-visual feedbacks spark the immersants' imagination, fostering whole-body engagement and intercorporeal collaboration with fellow immersants.

Live-coders dynamically creates metaphorical AVVM based on intercorporeal signals. AVVMs are patterns or structures within the GravField system that respond to specific intercorporeal signals derived from participants' movements. These signals include factors such as distance, area formation, relative height differences, and head motion synchronization. Participants will be introduced to the AVVMs developed thus far, such as the rope, spring, and magnetic field patterns.

The Rope: The rope AVVM encourages participants to swing collaboratively to gain maximum feedback. By synchronizing their swinging motions, participants create a continuous and rhythmic audiovisual experience. The rope pattern emphasizes the importance of coordination and shared movement, allowing participants to explore the concepts of rhythm, synchronization, and collective expression. Participants will engage in swinging movements together, experiencing the immersive and captivating nature of the rope pattern in generating collaborative music.

The Spring: The spring AVVM connects bodies with strong auditory feedback, encouraging participants to play with the distance between them. As participants move closer together, the auditory feedback strengthens, reinforcing their proximity and fostering a sense of unity. Conversely, when participants move apart, the auditory feedback diminishes, prompting them to explore the musical possibilities of distance. Participants will experiment with the spring pattern, exploring the relationship between distance and sound, and discovering how it influences their collaborative music creation.

The Magnetic Field: The magnetic field AVVM is designed to attract neighboring body relationships. By creating a spatial magnetic vector field between participants, it encourages individual exploration and expression while maintaining a sense of proximity and collaboration. Participants will learn how to utilize the magnetic field pattern to shape the dynamics and interactions between their bodies, creating distinct musical outcomes.

By engaging with these AVVMs, immersants explore the dynamics and interactions between their bodies, ultimately contributing to the collaborative music creation.

3 ARTIST INFORMATION

Botao 'Amber' Hu is a social computing researcher and experiential futures designer. He directs Reality Design Lab (<https://reality.design>), an independent interdisciplinary research and design lab exploring the intersection of soma design (bodily play), speculative design (artificial life), spatial computing (mixed reality), and social computing (protocol science, public-interest cryptography, decentralized AI governance). His primary focus is designing socially experiential futures using mixed reality as the main medium. He also serves as a visiting lecturer at the China Academy of Art. He holds a bachelor's degree in computer science from Tsinghua University and a master's degree in computer science with AI concentration from Stanford University.

Rem RunGu Lin is a digital artist, researcher, and technologist with a master's degree from MIT, currently pursuing a PhD in Computational Media and Art at HKUST (GZ). His work explores the intersections of embodied interaction, brainmedia, and media theory, blending research with creative practice. His current research focuses on three key areas: Media as Art, exploring how media theory can be applied to art and design to augment creative practices; Human as Media, investigating the entanglement of body and mind and how this relationship can be expressed through digital art; and Space as Media, extending the concept of human-machine-space symbiosis to create immersive, interactive bio-digital spatial entities.

Yuemin Huang is a Shanghai-born artist and researcher interested in how culture-constructed media shapes cognition, engaging with themes such as the materiality of digital infrastructure, polycentric self-organizing structures, artistic/technical protocols and constructible perception. Yuemin is also a co-founder of TINAG, an art game group, focusing on experimenting with different organizational structures through games, creating shared experiences within communities, and exploring the alternative reality gaming method. With works appearing in festivals, distributed papers and alternative written forms, including SeP Shanghai(2023), Hastac(2023), ICA(2023). Yuemin is a lecturer at East China Normal University. She holds a bachelor's degree in Public Art from East China Normal University and a master's degree in Art and Technology Studies from the School of the Art Institute of Chicago.

Mingze Chai is a sound designer, music editor, and sound artist, primarily focusing on audio design within immersive interactive experiences. As a Senior Audio Designer at Bilibili Games, Mingze played a key role in the interactive music system and sound design for video game titles such as 'Warm Snow', 'Soda Crisis', 'Higan: Eruthyll', and 'Thrud'. Deeply interested in the intersection of audio artistry and interactive design, his work continuously explores the potential of sound in elevating user experiences.

Xiaobo Aaron Hu is a creative technologist, VJ, and psychological researcher. His work focuses on the intersection of technology and psychology, developing new methods for psychological measurement and intervention. He also investigates how technological advancements impact societal psychology and moral ethics. Aaron's background includes leading the innovation lab at 4A Company, with over a decade of experience in new media, experiential marketing, and digital innovation. Skilled in both software and hardware, he excels at delivering optimal technological solutions.

Yilan Elan Tao is a designer and researcher specializing in HCI and Mixed Reality. Tao's exceptional design and presentation skills have been crucial to the success of Reality Design Lab's award-winning projects HoloKit, MOFA, GravField, etc. These projects have been showcased at prestigious conferences and venues such as CHI, SIGGRAPH, ISMAR, iF award, Red Dot Award, Core77 Awards, and SXSW. She holds a Master's degree in Architecture from Cornell University.